

Netflix

You are a Data Analyst on the Content Discovery Team at Netflix, tasked with evaluating the impact of the recommendation algorithm on user engagement....

Question 1: What is the total watch time for content after it was recommended to users? To correctly attribute watch time to the recommendation, it is critical to only include watch time after the recommendation was made to the user. A content could get recommended to a user multiple times. If so, we want to use the first date that the content was recommended to a user.

```
with calculate AS (  
  SELECT user_id, content_id, MIN(recommended_date) AS recommended_date  
  FROM fct_recommendations  
  GROUP BY user_id, content_id)  
SELECT SUM(w.watch_time_minutes) AS total_watch_time  
FROM fct_watch_history AS w  
JOIN calculate AS r  
  ON w.user_id = r.user_id AND w.content_id = r.content_id  
WHERE w.watch_date >= r.recommended_date;
```



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Question 2: The team wants to know the total watch time for each genre in first quarter of 2024, split by whether or not the content was recommended vs. non-recommended to a user.

Watch time should be bucketed into 'Recommended' by joining on both user and content, regardless of when they watched it vs. when they received the recommendation.

```
with calculate AS(
  SELECT genre, watch_time_minutes,
    CASE WHEN recommended_date IS NULL THEN 'Non-Recommended'
    ELSE 'Recommended' END AS Status
FROM fct_watch_history AS w
INNER JOIN dim_content AS d
ON w.content_id = d.content_id
LEFT JOIN fct_recommendations AS r
ON w.user_id = r.user_id AND w.content_id = r.content_id
WHERE TO_CHAR(watch_date, 'YYYY-MM') IN ('2024-01', '2024-02', '2024-03'))

SELECT genre, Status, SUM(watch_time_minutes) AS total_times
FROM calculate
GROUP BY genre, Status
```



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Question 3: The team aims to categorize user watch sessions into 'Short', 'Medium', or 'Long' based on watch time for recommended content to identify engagement patterns. 'Short' for less than 60 minutes, 'Medium' for 60 to 120 minutes, and 'Long' for more than 120 minutes. Can you classify and count the sessions in Q1 2024 accordingly?

```
with calculate AS(  
    SELECT watch_time_minutes,  
           CASE WHEN watch_time_minutes < 60 THEN 'Short'  
                WHEN watch_time_minutes > 120 THEN 'Long'  
                ELSE 'Medium' END AS Session_length  
FROM fct_watch_history AS w  
INNER JOIN fct_recommendations AS r  
ON w.user_id = r.user_id AND w.content_id = r.content_id  
WHERE watch_date BETWEEN '2024-01-01' AND '2024-03-31')  
  
SELECT session_length, COUNT(*) AS number_count  
FROM calculate  
GROUP BY session_length
```

